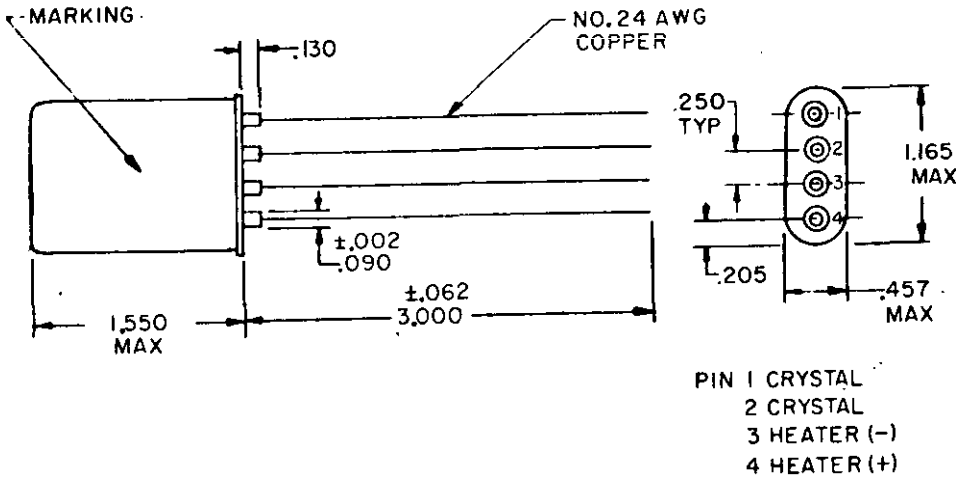
MILITARY SPECIFICATION SHEET

OVEN, CRYSTAL (+75°C)

The complete requirements for procuring the oven described herein shall consist of this document and the latest issue of specification MIL-O-39021.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Pertinent characteristics: 1 crystal, HC-25/U or HC-29/U, 5 watts, hermetic seal



PIN 1 CRYSTAL
2 CRYSTAL
3 HEATER (−)
4 HEATER (+)

INCH	MM
.002	.05
.062	1.57
.090	2.29
.130	3.30
.205	5.21
.250	6.35
.457	11.61
1.165	29.59
1.550	39.37
3.000	76.20

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm).
4. Case and base are supplied unassembled.
5. Designed for solder seal.
6. For regular and testing purposes, the units will be assembled.

FSC 5955

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REQUIREMENTS:

Materials:
Case: Nickel silver per QQ-C-585 or nickel plated steel per QQ-S-698.
Base: Type MFE, MIL-M-14.

Design and configuration: See figure.

Physical characteristics:
Weight: 1.5 ounces maximum.

Oven cavity: Shall accommodate one crystal unit in either a HC-25/U or HC-29/U (or equivalent) holder.

Grounding and circuit continuity:
See figure.

Terminal configuration and identification:
See figure.

Rated voltage: See table I.

Capacitance: 2 pF maximum.

Warmup power (oven heater elements):
Conditioning prior to test: 1 hour at -55° ±2°C.
Power: Not to exceed 5 watts.

Warmup time:
15 minutes maximum at -55°C, +25°C, and +75°C ambient temperatures.

Average operating power (oven heater element):
After warmup: Not to exceed 5 watts at 25°C.

Oven cavity temperature:
Shall be 75°C ±2°C at +25°C ambient.
Stabilization temperatures: +25°C, -55°C, +25°C, and +70°C. Tolerance for -55°C through +70°C shall be ±1°C from oven cavity temperature measured at +25°C ambient.

Cycling variation: ±.5°C.

Explosion: Not applicable.

Part number:
M39021/16-(dash number from table I).

TABLE I. Dash number and rated voltage.

Dash number	Rated voltage
01	24 to 28 Vdc
02	115 Vac ±5%
Solid state proportional control	
03	24 to 28 Vdc
04	115 Vac ±5%

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, SM
Navy - SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, OS, CG
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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